Data Cleaning, Transformation and Modelling

Question 1: What is Data Loading in Power BI and why is it considered the first step of analysis?

Data Loading is the process of connecting Power BI to a specific data source (such as Excel, SQL databases, or web links) and bringing that data into the Power BI environment.

- **Foundation of Analysis:** It is the first step because analysis cannot occur without a data substrate. You must establish a connection to define the scope of the information available.
- **Schema Definition:** During loading, Power BI begins to identify data types (Text, Integers, Dates), which determines how the software will allow you to interact with that data later.
- **Initial Extraction:** It serves as the "Extract" portion of the ETL (Extract, Transform, Load) process, ensuring the raw material is present for subsequent cleaning and modeling.

Question 2: Explain the difference between "Load" and "Transform Data" in Power BI.

When you connect to a dataset like the **Corona Virus 2020** data, Power BI offers two primary paths:

Feature	**Load**	**Transform Data**
Action	Imports data directly into the Power BI data model without changes.	Opens the **Power Query Editor** for data manipulation.
Use Case	Used when the data is already "clean" and requires no structural changes.	Used to remove duplicates, fix null values, or change data types.

Feature	Load	Transform Data
Data Quality	Assumes the source data is perfect.	Allows you to "shape" the data to meet specific analytical needs.
Outcome	Data is immediately available for visualization.	Data is processed through "Applied Steps" before entering the model.

Question 3: What is a Fact Table and a Dimension Table? Give examples from the dataset.

Effective data modeling relies on separating "events" from "descriptions":

- **Fact Table:** This contains the quantitative, measurable data (the "numbers"). It usually consists of foreign keys and aggregated values.
    - *Example (Corona Dataset):* A table containing the daily count of **Confirmed Cases**, **Deaths**, and **Recovery Rates**.
- **Dimension Table:** This contains the descriptive attributes (the "context") that filter and group the facts.
    - *Example (Corona Dataset):* A table containing **Country**, **State**, and **Date** information, or a table detailing **Vaccination Status** category.

Question 4: Why is Star Schema preferred over Snowflake Schema in Power BI?

While both are methods of organizing tables, the **Star Schema** is the gold standard for Power BI performance.

- **Star Schema:** Fact tables are connected directly to dimension tables. It is simpler and requires fewer "joins".
- **Snowflake Schema:** Dimension tables are "normalized" (split into further sub-tables), making it more complex.

Reasons for Preference:

1. **Performance:** Power BI is optimized for the Star Schema; fewer table relationships mean faster report rendering and quicker DAX calculations.

2. **Usability:** It is easier for users to navigate the "Fields" pane because the relationships are straightforward and less nested.
3. **DAX Efficiency:** Formulas written in DAX (Data Analysis Expressions) are simpler to write and less prone to errors when the model is not overly normalized.